

① Geometric distribⁿ: Gives the prob. of getting first success in k^{th} trial.

$$P_X(K) = (1-p)^{K-1} \cdot p \quad \text{PMF}$$

ex ①: Suppose basketball player makes a free throw with prob $p=0.3$, we want to model number of attempts until the first success.

$$\rightarrow P_X(K) = (1-p)^{K-1} \cdot p \quad K = 1, 2, \dots, n, \dots$$

$$P_X(1) = (1-0.3)^{1-1} \cdot 0.3 = 0.3$$

prob. of getting first success in 1st trial

$$\begin{aligned} P_X(x=3) &= (1-0.3)^{3-1} (0.3) \\ &= (0.7)^2 (0.3) \rightarrow \text{prob of} \\ &\text{getting first success in the 3rd trial.} \end{aligned}$$

Poisson distributⁿ. A factory produces ⁿ1000 light bulbs per day, and on an average 5 of them are defective. Now, we want to find out probability of getting exactly 3 defective bulbs in a day.

$\lambda = 5$ (average rate of defective bulbs).

$$P_x(k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k=0, 1, \dots$$

$$P_x(3) = \frac{e^{-5} (5)^3}{3!} = 0.1404$$

↓

this is prob. of getting
3 defective bulbs in a day.